Campus Information Retrieval Assistant Based on SLM

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ABSTRACT

Universities keep important information like the details of faculty members, policies, campus facilities, administrative guidelines, etc. on diverse platforms like websites, pdfs, and notices. This dispersed distribution is making it hard for students and stakeholders to find reliable information fast. This project calls for a Campus Information Retrieval Assistant built on a Small Language Model (SLM) coupled with Retrieval-Augmented Generation (RAG). The system extracts relevant information from the trusted sources of the institutions, and creates responses based on the verified information on the campus. By working on the local infrastructure, the assistant guarantees privacy and dependence on cloud-based models and delivers fast and accurate responses to queries concerning the campus.

INTRODUCTION

Accessing university information can be often time consuming due to the variety of platforms and documents it is spread across. Students often need to know information related to faculty, facilities on campus, grading systems, or academic policies, but searching for this information manually may be inefficient. Many modern chatbot systems are based on large language models that can deliver incorrect responses and raise privacy concerns due to the fact that they are cloud-based.

To get over these challenges, this project presents a Campus Information Retrieval Assistant based on a Small Language Model and Retrieval-Augmented Generation. The system gathers data from institutional sources and forms answers under the basis of proven data. By prioritizing the campus documents and operating locally, the system helps to increase the reliability, privacy and accessibility of university information.

OBJECTIVES

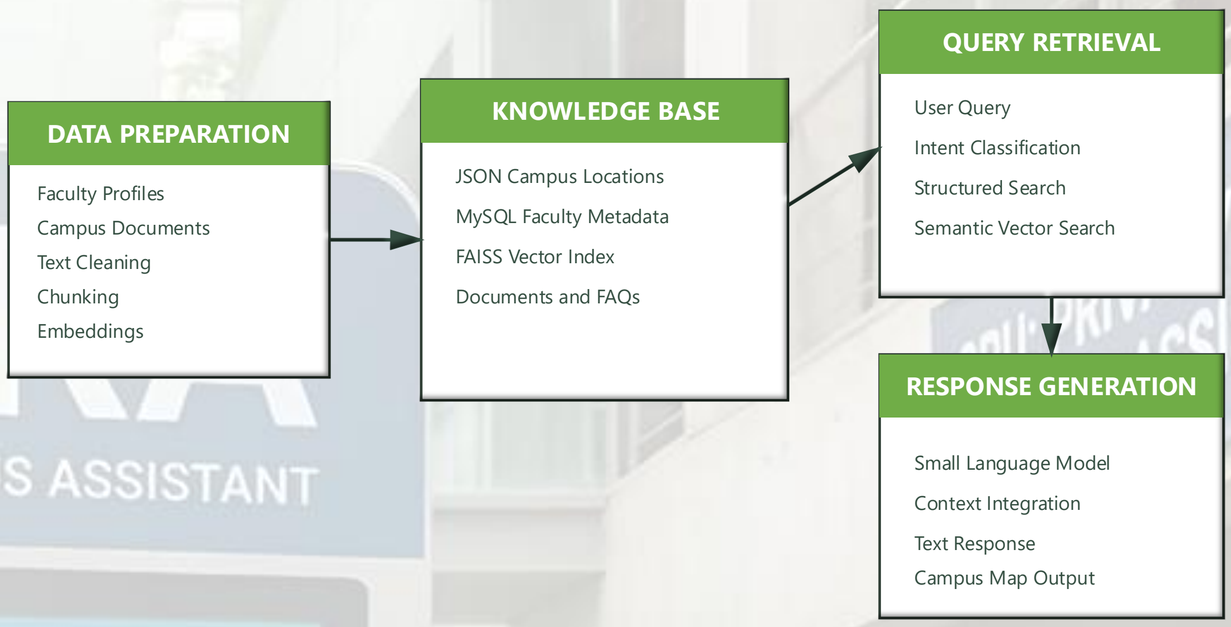
i. To design an information retrieval assistant specific to the campus using a Small Language Model.
ii. To integrate Retrieval-Augmented Generation to give Accurate and Grounded Response
iii. In order to prioritize official institutional data sources for reliable answers.
iv. To develop capabilities for natural language interaction for students and stakeholders of the university.
v. To support on-premises deployment for privacy and efficiency

METHODOLOGY

The system is initialized to gather information related to the campus like details of faculty members, academic policies, campus facilities, and frequently asked questions from official sources. The data collected is subsequently cleaned and broken into smaller segments of the text in order to enhance data retrieval performance.

These chunks of text are transformed into vector embeddings which are saved in a vector database for similarity search. When a user submits a query, the most relevant information is retrieved from the system based on the semantic similarity. The context retrieved is then fed to the Small Language Model, which responds based on the context retrieved. A web-based chat-interface enables users to interact with the assistant and get campus information in a short time.

SYSTEM WORKFLOW



CONCLUSION

The proposed Campus Information Retrieval Assistant shows an effective solution to the problem of enhancing access to university information. By using a Small Language Model and Retrieval-Augmented Generation, the system provides accurate responses from reliable institutional data. The usage of the local deployment enhances privacy and lessens the dependency on huge cloud-based models. This way, it improves the accessibility of the information from the campus and may be further expanded for real-life deployment to the schools.

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